NoLand Engineering, Inc.

MODEL RS11 v4
CANbus Engine Data Converter

This manual covers **version 4.0**. Future versions may contain enhancements not covered herein.

TABLE OF CONTENTS

Introduction
Installation
Analog Inputs
RPM Connection
CANbus Interface
USB Interface
Setup Utility
Technical Specifications
Warranty

TABLES & FIGURES

Figure 1 - RS11 Installation Examples
Figure 2 - Twin RS11 Installation
Figure 3 - Analog (gauge) connections
Figure 4 - RPM connections
Figure 5 - Setup Utility
Specifications

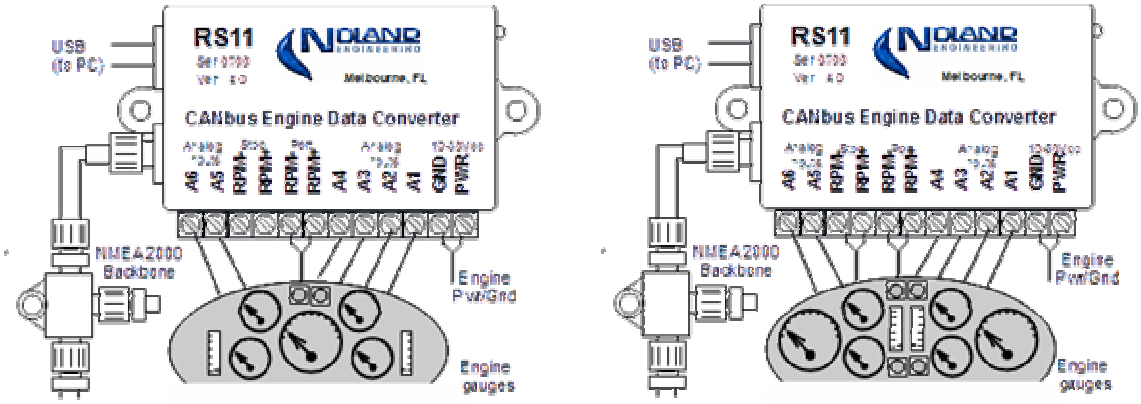
Introduction: The RS11 converts tachometer signals and analog gauge voltages from virtually any engine into CANbus digital messages compatible with J-1939 and NMEA 2000(N2K) protocols. It is configurable for either single or dual engine installations. The user must setup the RS11 for the desired data and calibrate pulses/rev, analog scale factors, etc. Configuration is performed via a USB PC interface which also serves as an output data monitor, if desired. Output messages are applied to the CANbus, which can then be viewed by any compatible display.

The RS11 is designed for versatile interfacing to almost any type of analog engine sensor or gauge commonly used on marine or automotive engines. It has two RPM (Tach) inputs, six analog(Gauge) inputs and supports both 12 and 24 volt systems. The analog inputs can all be assigned to one engine or split between two engines. Multiple RS11's can be installed on the same CANbus.

Although intended for direct connection to existing gauges, the RS11 can be used with resistive senders, where gauges have been eliminated. In such cases “Sender Current” is provided by the RS11 when enabled by the user.

The user must perform an initial configuration of the unit to suit the particular engine(s). This consists of selecting the specific information to be converted and entering proper calibration factors. A PC–based Setup Utility is included to simplify configuration and calibration .

Installation: Typical installations for a single and dual engine vessel are shown in Figure 1. For the single engine case, only the 'Port' RPM input is used, and all analog inputs are available for that engine. In the dual engine case, the analog inputs must be divided between the two engines. This usually means each engine only gets three analog inputs. To get six analog inputs to each of two engines, you can use two RS11's, one for Port and another for Starboard Instance (see Figure 2). Note that the RPM inputs are specifically assigned to Port and Starboard, whereas the Analog inputs can be independently assigned to Port or Starboard, as desired.



Single engine

Dual engine

Figure 1. Installation Examples

All engine sensor and/or gauge connections are made to a terminal strip on the RS11. CANbus connection is via the 5-pin "Micro-C" connector. A USB PC cable is supplied for configuring the RS11, but may be left in place for data monitoring (via PC) if desired.

The RS11 needs engine power to operate, which is usually the ships' batteries. Furthermore, it is desirable to have the RS11 power shut off whenever the engines are shut off, so as not to trigger alarms on your displays.

The individual interfaces (CAN, USB, Engine) are all galvanically isolated from each other to prevent "ground" conflicts. The user must connect an engine power source (10-30 Vdc) to the RS11 before it functions. USB and NMEA 2000 interfaces are powered by their own respective interfaces, with full data isolation. This means that that your engines can be powered by the ship's batteries, your PC by an AC generator and your CANbus by a separate isolated supply with no power, ground or data conflicts.

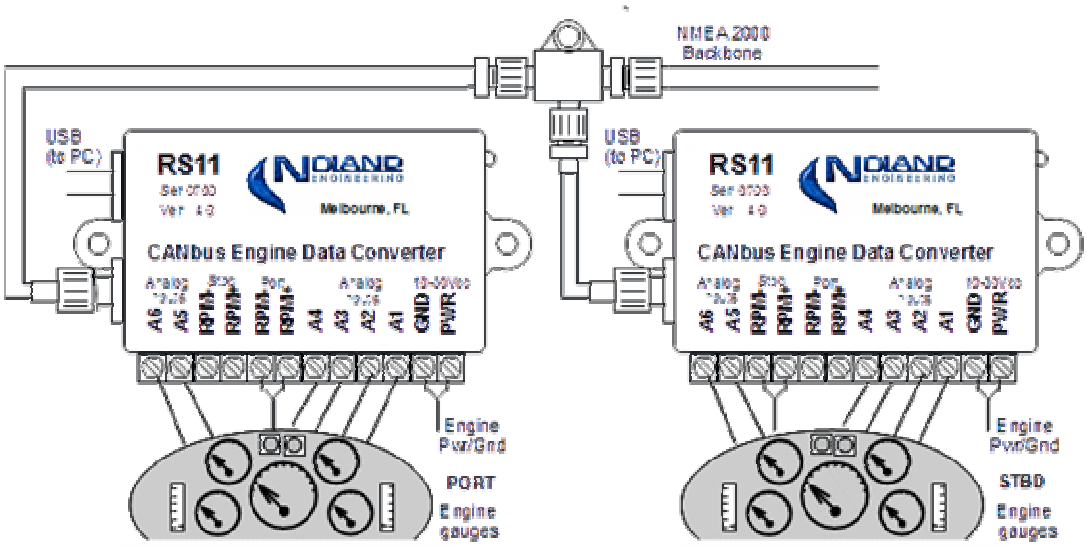
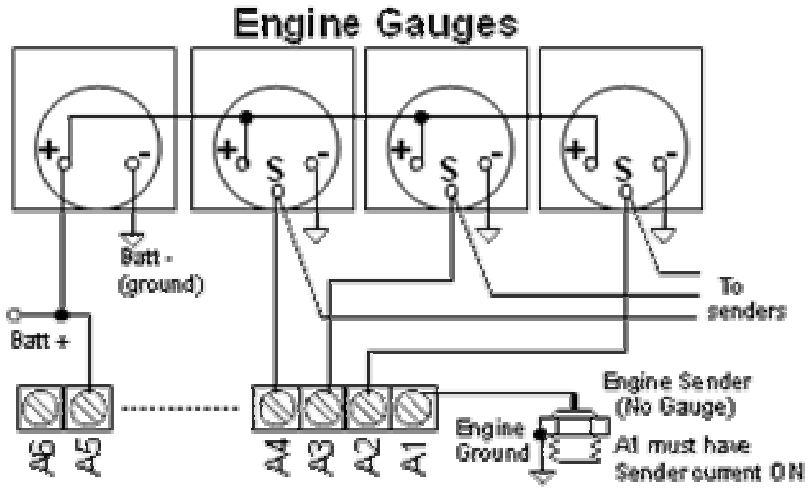


Figure 2. Twin RS11 Installation

<u>Analog (Gauge) inputs</u>: The RS11 A1-A4 inputs accept voltages from 0-20V while A5,A6 accept 0-30V. Normally these connect directly to existing gauges or voltage-output sensors, but A1-A4, can also measure resistive senders directly when no gauges are present. The user must enable internal current sources for this function. Inputs A5 and A6 do not have current sources but can be used for direct 'Battery Volts' measurement up to 30V. They can also be used for other measurements (pressure, temp., …) when gauges are present.

Figure 3 shows typical connections. For existing gauges, the 'S' terminal on the back of each gauge is where the connection should be made. For no gauge, the connection is as shown on the 'A1' terminal with the 'Sender Current' for 'A1' turned on. Battery voltage is measured directly by A5 and/or A6.

Figure 3. Analog (Gauge) Connections



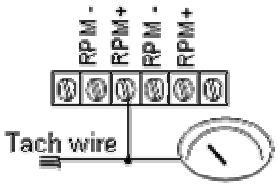
RPM (Tach) Connection: RPM signals come from different sources depending on the engine. They may come from an alternator output, the ignition coil, an ECU (engine control unit), or some type of AC or pulse sender (diesel engines). The RS11 will interface to all of these, but the connection details vary by type. The RS11 sends 'Engine Hours' whenever RPM is > 100.

Figure 4a. shows how to connect the RS11 to an ignition coil or alternator output signal. Since there is only one Tach signal wire, it must connect to the 'RPM+' terminal. The 'RPM-' terminal should usually be left disconnected (open).

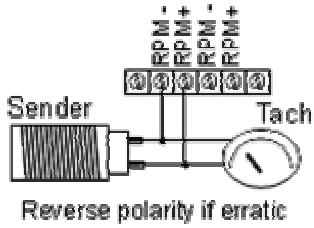
Figure 4b. shows how to connect to a sender or generator used on many diesel engines. These senders have 2 wires going to the Tachometer , which should be connected to the 'RPM+/ RPM-' terminals of the RS11. If the RPM reading is not stable, the polarity may need to be reversed.

Figure 4c. shows an alternative connection, which may work better with some senders, especially if one wire of a 2-wire sender is grounded. The ungrounded wire of the sender should go to 'RPM+' while the 'RPM-' terminal is grounded. This connection increases the input sensitivity, but is only used when necessary.

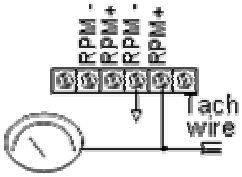
Figure 4. RPM Connections



a) Most single wire Tachs



b) Most two wire Tachs



c) Some Tachs, including 2-wire senders with a wire grounded.

CANbus Interface: The primary purpose of the CANbus interface is to output the acquired engine data to appropriate Multi-function Displays. Connection to the bus is via a 'Micro C' connector on the side of the RS11.

Setting up the RS11 for N2K output is done by the user. A software program called "RS11 Setup Utility" is included for this purpose. A Windows PC with USB is needed to run this utility. The available assignments for each analog input are shown in Table 1. Default assignments for each input are in gray, and alternative assignments are listed below each default.

Figures 1 and 2 show typical connections to an NMEA 2000 CANbus backbone. The backbone consists of cables and Tee's which supply both DC power (8-16V) and a data interface to each device. The CANbus must have at least one 120 ohm termination, and very long buses should have a termination at each end.

A1 (0-20V)	**A2** (0-20V)	**A3** (0-20V)	**A4** (0-20V)	**A5** (0-30V)	**A6** (0-30V)
Turbo (P)	Tilt/Trim (P)	Turbo (S)	Tilt/Trim (S)	Batt Volt(P)	Batt Volt(S)
Oil Press	Oil Press	Oil Press	Oil Press	Oil Press	Oil Press
Oil Temp	Oil Temp	Oil Temp	Oil Temp	Oil Temp	Oil Temp
Cool Temp	Cool Temp	Cool Temp	Cool Temp	Cool Temp	Cool Temp
Cool Press	Cool Press	Cool Press	Cool Press	Cool Press	Cool Press
Fuel Press	Fuel Press	Fuel Press	Fuel Press	Fuel Press	Fuel Press
Fuel Level	Fuel Level	Fuel Level	Fuel Level	Fuel Level	Fuel Level
Tran Press	Tran Press	Tran Press	Tran Press	Tran Press	Tran Press
Tran Temp	Tran Temp	Tran Temp	Tran Temp	Tran Temp	Tran Temp
Batt Amps	Batt Amps	Batt Amps	Batt Amps	Batt Amps	Batt Amps

USB Interface: The USB interface is used for setup and calibration of the RS11. It also allows monitoring of the RS11 and updating firmware, when necessary.

When the USB interface is connected to a PC, a 'Virtual COM Port' is created within the PC, which is used by any PC communicating with th RS11. This happens even if the RS11 has no DC power. USB Drivers for the RS11are included on the "Installation CD".

To install the RS11 USB drivers, do the following:
1) Plug RS11 into USB port of computer
2) Insert RS11 Installation CD into disk drive of computer
3) When New Hardware Wizard appears, select 'Windows recommended option'
4) Click "Next"
5) Click "Finish"
If your computer does not automatically install the "USB Port Drivers," a "New Hardware Wizard" will appear. In this case:
1) When New Hardware Wizard appears, select 'Windows recommended option'
2) Click "Next"
3) Click "Finish"

RS11 Setup Utility: This PC based utility (Figure 5) is used to configure and calibrate the RS11 to send NMEA 2000 messages from the incoming analog and RPM engine data. The utility allows you to select the data type for each analog input and enter calibration data for all inputs. It need only be used once with the RS11 unless changes have to be made later on. It is included on the CD disk supplied with the RS11.

The RS11 connects to this utility through a USB Virtual COM Port. It is interactive in that the Utility can both send and read from the RS11, so that proper configuration can be verified. Most configuration consists of selecting options from lists or entering numbers in boxes.

Calibration of analog voltages is made simpler because the RS11 provides a direct reading of the voltage on each analog input. Thus, a voltmeter is not needed. Details of how to use the Setup Utility are contained within the Utility's "Help" section.

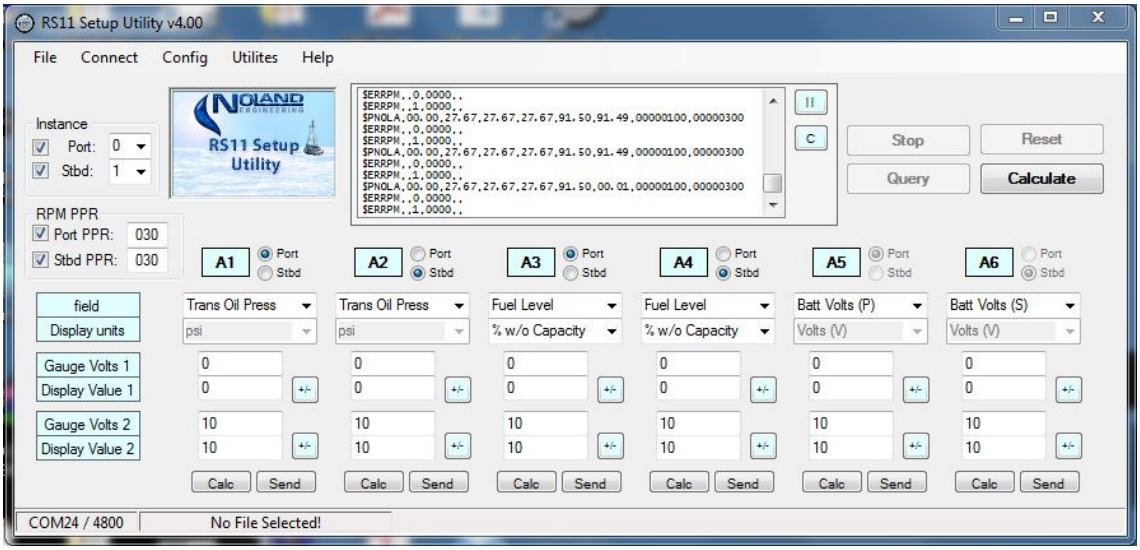


Figure 5. Setup Utility

TECHNICAL SPECIFICATIONS:

RPM resolution	+/-2 (ppr=200), +/-4 (ppr=60), +/-16 (ppr=12)
Tach input impedance	> 100 Kohm
Analog input accuracy	+/-2%
Analog input impedance	12 Kohm
Power	10-30 Vdc, 50ma. Max
Temperature, Humidity	0-50 degC, 0-100% (non-condensing)
Size, weight	2.0” x 3.4” x 0.9”, 3 oz.

Data PGN’s (NMEA 2000)

127488	Engine Parameters, Rapid Update
127489	Engine Parameters, Dynamic
127493	Transmission Parameters, Dynamic
127503	Fluid (Fuel) Level
127508	Battery Status

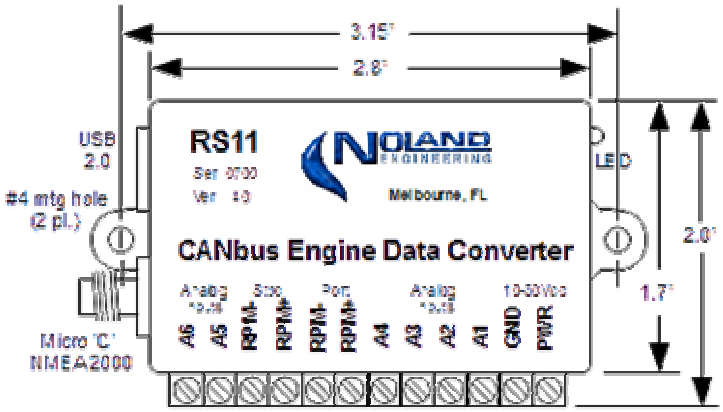
WARRANTY: Subject to the terms, conditions and limitations herein, NoLand Engineering warranties this product against manufacturing and material defects for a period of 24 months from the date of purchase. NoLand Engineering, at its discretion, will repair or replace the defective unit with new or refurbished parts. There is no other express warranty or representation of any kind that we make concerning this product. Such repair or replacement is the sole remedy to the customer, consistent with local and provincial laws. Any repaired or replaced unit will be warrantied for the balance of the original product's warranty period.

It is customer's responsibility to return products safely to NoLand Engineering. Unless other arrangements are made, the customer must properly pack, ship, and insure the unit for return. If proof of purchase is not provided by the customer, NoLand Engineering will use its own records to determine the date of purchase.

This Limited Warranty does not cover the following:

- products with alterations to the serial number, mutilated labels, or missing labels
- failure from abuse, misuse, accident, over-voltage, unauthorized repair or modification, or improper installation.
- costs incurred from re-installation and testing.

This Warranty gives you specific legal rights, and other rights, which vary from state to state.



NOLAND ENGINEERING
728 E Lincoln Ave., #3
Melbourne FL 32901

PH:321 951-7329 FAX:321 951-8773
http: // www . nolandeng . com
email: *info@nolandeng.com*